Document 06

Mapping Canon

1. Purpose of This Document

This document defines the authoritative mapping assets and lookup structures for Trinity Engine v1.

Its purpose is to specify:

- the canonical zodiac sign mapping basis where needed for output assignment
- the canonical Human Design gate order mapping
- the canonical line subdivision basis
- the canonical channel table
- the canonical center list
- the canonical channel to center mapping
- the canonical Incarnation Cross structural mapping basis where applicable
- versioning requirements for all mapping assets

This document defines what lookup structures are authoritative.
It does not replace the Calculation Canon. It supports the Calculation Canon.

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2. General Mapping Principles

2.1 Mapping Authority Rule

Only mappings explicitly pinned in this document are authoritative for Trinity Engine v1.

No mapping may be inferred from:

- community convention
- external calculators
- unofficial diagrams
- prior prototype code
- library defaults
- remembered Human Design gate sequences
- remembered channel lists

If a mapping is required by calculation but not pinned here, that is a canon gap.

2.2 Mapping Version Rule

All mapping assets in Trinity Engine v1 must be versioned.

At minimum, the final engine output metadata must reference:

- `mapping_version`

If separate mapping groups are versioned independently, the canon must either:

- expose separate mapping version identifiers
  or
- define one unified mapping version covering all pinned mapping assets

For v1, a single unified `mapping_version` is preferred unless there is a strong operational reason to split it.

2.3 Deterministic Lookup Rule

All lookup operations based on this document must be deterministic.

This means:

- one input maps to one output
- no table may contain ambiguous duplicates
- no table may contain overlapping interval ownership
- no lookup may require interpretation at runtime

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3. Astrology Mapping Canon

3.1 Tropical Zodiac Sign Order

The authoritative zodiac sign order for Trinity Engine v1 is:

1. Aries
2. Taurus
3. Gemini
4. Cancer
5. Leo
6. Virgo
7. Libra
8. Scorpio
9. Sagittarius
10. Capricorn
11. Aquarius
12. Pisces

This order is authoritative for astrology sign placement.

3.2 Zodiac Sign Interval Basis

Astrology sign placement must use the tropical zodiac division of the full 360 degree circle into 12 equal sign intervals of 30 degrees each.

The canonical interval ownership rule is:

- interval start inclusive
- interval end exclusive

Examples:

- Aries = `[0.0, 30.0)`
- Taurus = `[30.0, 60.0)`
- Pisces = `[330.0, 360.0)`

A longitude of exactly `30.0` belongs to Taurus, not Aries.

This rule must remain consistent with the longitude normalization rule in Document 05.

3.3 Astrology Object Identifier Set

The authoritative astrology object identifier set for v1 is:

- `sun`
- `moon`
- `mercury`
- `venus`
- `mars`
- `jupiter`
- `saturn`
- `uranus`
- `neptune`
- `pluto`
- `chiron`
- `north_node_mean`
- `earth`

These identifiers are authoritative for structured output and internal mapping references.

No additional astrology object identifier is in scope unless added by later canon revision.

3.4 House Identifier Set

The authoritative house identifiers for v1 are:

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11
- 12

These identifiers apply to:

- house cusp outputs
- house placement outputs

No alternate house label scheme is authoritative unless explicitly added in the Output Specification.

House interval ownership is not defined by this section.
The authoritative house interval ownership rule for house placement is defined in Document 05.

3.5 House Placement Dependency Note

Where astrology house placement is required, Document 06 provides only the canonical house identifier set.

The authoritative house placement boundary behavior, including cusp ownership and wraparound behavior between House 12 and House 1, is defined exclusively in Document 05.

No implementation may infer house interval ownership from:

- identifier numbering
- external astrology libraries
- unofficial house conventions
- assumptions derived from sign interval rules

4. Human Design Gate and Line Mapping Canon

4.1 Mandala Start Point

The authoritative Human Design mandala anchor for Trinity Engine v1 is:

- the canonical gate sequence begins with **Gate 38** at **277.5°**
- the canonical Gate 41 segment begins at **300.0°**

This anchor is adopted because it is consistent with:

- the official Human Design principle that the Rave New Year begins when the Sun enters Gate 41, usually on January 21 to 22
- the tropical zodiac boundary where Aquarius begins around January 20 to 21, corresponding to 300.0° ecliptic longitude
- the pinned 64 gate transit sequence used elsewhere in this canon package

Any superseded or previously rejected anchor assumption is non authoritative and must not be used in Trinity Engine v1.

4.2 Gate Width

The authoritative gate width is:

- `5.625°`

This is equal to:

- `360° / 64`

Every gate interval in the mandala occupies exactly one gate width.

4.3 Line Width

The authoritative line width is:

- `0.9375°`

This is equal to:

- `5.625° / 6`

Every gate interval is subdivided into exactly 6 equal line intervals.

4.4 Interval Ownership Rule

Human Design gate and line intervals are half open intervals:

- interval start inclusive

- interval end exclusive

This rule is authoritative for:

- gate assignment
- line assignment
- boundary handling
- wraparound handling

A longitude exactly on an interval boundary belongs to the next segment if that boundary is the previous segment's end.

4.4A Mapping and Calculation Boundary Rule

The interval ownership rules and mapping assets defined in this document are authoritative only for the mapping domains explicitly covered here.

Document 06 does not override:

- longitude normalization rules defined in Document 05
- Design time derivation rules defined in Document 05
- Design time final timestamp selection and whole second emission rules defined in Document 05
- output formatting and comparison rules defined in Document 07 and Document 10

Where a mapping asset in this document is consumed by a calculation defined elsewhere in the canon, the governing calculation rule remains authoritative.

4.5 Canonical Gate Order List

4.5.1 Purpose of This Section

This section defines the authoritative fixed 64 gate order sequence for the Human Design mandala in Trinity Engine v1.

This gate order list is required for deterministic gate assignment and must be treated as a canonical lookup asset, not as a procedurally inferred sequence.

4.5.2 Basis of This Section

The gate order list below is derived from:

- the official Human Design principle that the Rave New Year begins when the Sun enters Gate 41 in late January.
- a continuous month by month Human Design transit sequence that remains internally consistent across the annual cycle and returns to Gate 38 after Gate 58 without duplication before completing the loop.

This section supersedes any previously used or rejected alternative gate sequence anchor assumption.

4.5.3 Canonical Gate Sequence Rule

The authoritative Human Design gate order list for the 64 sequential gate segments on the mandala is:

1. 38
2. 54
3. 61
4. 60
5. 41
6. 19
7. 13
8. 49
9. 30
10. 55
11. 37
12. 63
13. 22
14. 36
15. 25
16. 17
17. 21
18. 51
19. 42
20. 3
21. 27
22. 24
23. 2
24. 23
25. 8
26. 20
27. 16
28. 35
29. 45
30. 12
31. 15
32. 52
33. 39
34. 53
35. 62
36. 56
37. 31
38. 33
39. 7
40. 4

41. 29
42. 59
43. 40
44. 64
45. 47
46. 6
47. 46
48. 18
49. 48
50. 57
51. 32
52. 50
53. 28
54. 44
55. 1
56. 43
57. 14
58. 34
59. 9
60. 5
61. 26
62. 11
63. 10
64. 58

After gate 58, the sequence returns to gate 38 and repeats cyclically.

4.5.4 Adoption Rule

The gate sequence listed in Section 4.5.3 is adopted as the authoritative canonical 64 gate order list for Trinity Engine v1.

This sequence must be implemented exactly as written.

No engineer may:

- replace it with an alternative sequence from another source
- reorder it based on community convention
- infer a different sequence from external diagrams or calculators
- treat it as provisional unless the canon is formally revised

Any future change to this sequence requires:

- formal canon revision
- mapping version update
- regression update
- implementation update

4.5.5 Direction Rule

This gate sequence must be traversed in the same directional order used by the gate mapping logic defined in Document 05.

The first segment in the sequence corresponds to the canonical first gate interval immediately following the final chosen mandala anchor point.

No reverse traversal is allowed unless explicitly introduced by future canon revision.

4.5.6 Anchor Confirmation Rule

The canonical gate sequence in Section 4.5.3 is anchored as follows:

- Gate 38 begins at 277.5°
- Gate 41 begins at 300.0°

This anchor is authoritative for:

- relative mandala position derivation
- gate assignment
- line assignment
- downstream Human Design and Gene Keys structural calculation

The use of this anchor remains subject to the governing calculation mechanics defined in Document 05, including canonical longitude normalization and half open interval application.

No alternate anchor is authoritative in v1 unless introduced by formal canon revision.

4.5.7 Status Recommendation

Status: Final

The canonical gate sequence and its angular anchor are now pinned for Trinity Engine v1.

4.6 Gate Identifier Rule

Gate identifiers must use one canonical representation only.

For v1, the preferred representation is:

- integer gate numbers 1 through 64

No alternate naming scheme is authoritative unless explicitly added for output presentation outside the engine.

4.7 Line Identifier Set

The authoritative line identifier set is:

- `1`
- `2`
- `3`
- `4`
- `5`
- `6`

These identifiers apply to:

- Human Design line outputs
- Gene Keys line outputs
- Profile construction
- Incarnation Cross structural encoding where line values are included

No alternate line labeling system is authoritative for v1.

4.8 Human Design Object Identifier Set

The authoritative Human Design object identifier set for both Personality and Design snapshots is:

- `sun`
- `earth`
- `north_node`
- `south_node`
- `moon`
- `mercury`
- `venus`
- `mars`
- `jupiter`
- `saturn`
- `uranus`
- `neptune`
- `pluto`

This identifier set must be used consistently across:

- snapshot activation records
- Incarnation Cross encoding
- Gene Keys source mapping where applicable

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5. Human Design Structural Mapping Canon

5.1 Canonical Center List

The authoritative Human Design center list for Trinity Engine v1 is:

- `head`
- `ajna`
- `throat`
- `g`
- `ego`
- `solar_plexus`
- `sacral`
- `spleen`
- `root`

These nine centers are exhaustive for v1.

No alternate center naming scheme is authoritative unless separately specified for presentation only.

5.2 Canonical Channel Table

The authoritative Human Design channel table is the following explicit set of valid gate pairs.

Each entry defines:

- `channel_id`
- `gate_a`
- `gate_b`
- `center_a`
- `center_b`

A channel is valid only if it appears in this canonical table.

No gate pair may be treated as a valid channel merely because it appears in unofficial diagrams, community materials, or remembered Human Design knowledge.

channel_id	gate_a	gate_b	center_a	center_b
1-8	1	8	g	throat
2-14	2	14	g	sacral
3-60	3	60	sacral	root
4-63	4	63	ajna	head
5-15	5	15	sacral	g
6-59	6	59	solar_plexus	sacral
7-31	7	31	g	throat
9-52	9	52	sacral	root
10-20	10	20	g	throat
10-34	10	34	g	sacral
10-57	10	57	g	spleen
11-56	11	56	ajna	throat

12-22	12	22	throat	solar_plexus
13-33	13	33	g	throat
16-48	16	48	throat	spleen
17-62	17	62	ajna	throat
18-58	18	58	spleen	root
19-49	19	49	root	solar_plexus
20-34	20	34	throat	sacral
20-57	20	57	throat	spleen
21-45	21	45	ego	throat
23-43	23	43	throat	ajna
24-61	24	61	ajna	head
25-51	25	51	g	ego
26-44	26	44	ego	spleen

27-50	27	50	sacral	spleen
28-38	28	38	spleen	root
29-46	29	46	sacral	g
30-41	30	41	solar_plexus	root
32-54	32	54	spleen	root
34-57	34	57	sacral	spleen
35-36	35	36	throat	solar_plexus
37-40	37	40	solar_plexus	ego
39-55	39	55	root	solar_plexus
42-53	42	53	sacral	root
47-64	47	64	ajna	head

For v1, the preferred canonical channel identifier format is:

- ascending gate pair string format: `gate_low-gate_high`

5.3 Channel to Center Mapping Rule

Each canonical channel entry explicitly includes the two centers it connects.

For Trinity Engine v1, the canonical channel table in Section 5.2 is also the authoritative channel to center mapping asset.

The engine must not infer connected centers from memory, unofficial diagrams, or external references if the channel table itself already defines them.

No separate alternate channel to center mapping may override this table unless introduced by formal canon revision.

5.3A Channel Table Consumption Rule

The canonical channel table defined in Section 5.2 is a lookup asset used by the structural derivation rules in Document 05.

Document 06 defines which channel pairs and center connections are authoritative.
Document 05 defines when those mappings are considered active in calculation.

No implementation may infer channel activation solely from remembered Human Design relationships or alternate source material outside the combined activation logic defined in Document 05.

5.4 Motor Center Set

For v1, the authoritative motor center set is:

- `root`
- `sacral`
- `solar_plexus`
- `ego`

This set is authoritative for:

- Type derivation
- motor to Throat connectivity analysis
- Authority derivation where relevant

No other center is considered a motor center in v1.

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6. Definition and Connectivity Mapping Canon

6.1 Definition Connectivity Basis

Definition must be derived from the connectivity graph formed by:

- defined centers
- active channels from the canonical channel table

No alternate connectivity asset is needed if the channel table and center list are complete and explicit.

6.2 Supported Definition Class Set

The authoritative Definition class set for v1 is:

- `none`
- `single`
- `split`
- `triple_split`
- `quadruple_split`

No other Definition label is in scope unless added by later canon revision.

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7. Profile and Incarnation Cross Mapping Canon

7.1 Profile Format Basis

Profile is structurally derived from:

- Personality Sun line
- Design Sun line

The canonical format is:

- `personality_line/design_line`

Example:

- `3/5`

No alternate ordering is authoritative.

7.2 Incarnation Cross Structural Basis

Incarnation Cross is structurally derived from:

- Personality Sun gate and line
- Personality Earth gate and line
- Design Sun gate and line
- Design Earth gate and line

7.2.1 Naming Boundary

A human readable Incarnation Cross naming table is not required for v1.

Unless an explicit authoritative naming asset is pinned later, the structural four activation encoding remains the authoritative output representation.

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8. Gene Keys Mapping Canon

8.1 Gene Keys Identifier Basis

Gene Keys key identifiers must be identical to Human Design gate identifiers.

For v1:

- Human Design gate number = Gene Keys key number

No remapping layer exists between Human Design gates and Gene Keys identifiers in v1.

8.2 Gene Keys Line Basis

Gene Keys line identifiers must be identical to Human Design line identifiers.

For v1:

- Human Design line = Gene Keys line

No alternate line system exists for Gene Keys in v1.

8.3 Activation Sequence Position Mapping

The authoritative Activation Sequence position mapping is:

- `life_work` = Personality Sun
- `evolution` = Personality Earth
- `radiance` = Design Sun
- `purpose` = Design Earth

These identifiers are authoritative for structured engine output in v1.

No additional Gene Keys sequence position is in scope.

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9. Identifier Normalization Rules

9.1 Canonical Case Rule

All canonical identifiers defined in this document must use one stable case convention.

For v1, the preferred convention is:

- lowercase snake_case for named identifiers
- integers for gate and line numbers
- hyphen joined ascending gate pairs for channel identifiers

Examples:

- `solar_plexus`
- `north_node_mean`
- `life_work`
- `34-20`

9.2 No Alias Rule

The engine must not accept or emit multiple aliases for the same canonical identifier.

Examples of forbidden alias drift:

- `solar plexus` and `solar_plexus`
- `ego` and `will` as separate engine identifiers
- `north node` and `north_node`
- `g_center` and `g`

If presentation layers need display friendly labels, those belong outside the engine.

9.3 Ego Alias Rejection Rule

For Trinity Engine v1, the canonical center identifier is:

- `ego`

The label:

- `will`

is not a separate engine identifier and must not appear in canonical output.

If presentation layers want to display “Will,” that belongs outside the engine and must not change the canonical identifier.

9.4 No Cross Domain Alias Inference Rule

No canonical identifier, mapping asset, or interval rule defined in this document may be extended by analogy into another domain unless the relevant governing canon document explicitly says so.

Examples:

- zodiac sign interval rules do not define house interval ownership
- house identifier numbering does not define house cusp ownership
- gate interval ownership does not define Design time timestamp emission
- presentation aliases do not become engine identifiers

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10. Out of Scope Mapping Assets

The following are out of scope for Trinity Engine v1 unless explicitly added by later canon revision:

- Gene Keys shadow, gift, and essence dictionaries
- Human Design strategy labels
- Human Design not self and signature dictionaries
- Human Design Variables mapping assets
- arrows, digestion, cognition, environment, motivation, view, perspective mappings
- astrology interpretive keyword dictionaries
- multilingual display dictionaries inside the engine

These belong to the content or presentation layer, not the calculation layer.

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11. Required Final Mapping Assets Checklist

Document 06 cannot be considered Final until the following assets are explicitly present in this document:

1. final confirmed angular anchor for the canonical gate sequence
2. full Human Design channel table
3. full channel to center mapping, either embedded in the channel table or as a separate explicit table

If any of these are missing, Document 06 remains incomplete.